Executive Summary

System Capacity and Care Load Analytics for Unaccompanied Children (UAC): A Time Series Monitoring and Forecasting Framework

The Unaccompanied Alien Children (UAC) program operates as a multi-stage humanitarian care pipeline involving intake, custody, transfer, and discharge processes managed across multiple federal agencies. Effective system management requires continuous monitoring of care capacity, inflow-outflow balance, and operational load. However, traditional reporting systems often focus on descriptive summaries and lack structured analytics for identifying sustained stress periods or forecasting future system demand.

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This research proposes a **data-driven analytical framework** designed to transform raw operational data into actionable insights for proactive decision-making. The framework integrates **data validation, feature engineering, trend monitoring, stress detection, and time-series forecasting** to monitor capacity dynamics within the UAC system.

The dataset used in this study consists of daily operational records that describe the flow of children through the care pipeline. Key variables include CBP daily intake, total custody counts, transfers to HHS facilities, children currently under HHS care, and discharge counts. To ensure reliability, the dataset undergoes validation checks such as identifying missing reporting dates, detecting duplicates, and enforcing logical constraints between operational variables.

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To convert raw operational counts into interpretable metrics, several analytical indicators are engineered. These include **Total System Load**, representing the total number of children under system responsibility; **Net Daily Intake**, measuring the difference between transfers and discharges; and **Care Load Growth Rate**, capturing how system burden changes over time. Together, these indicators provide a structured view of system capacity pressure and operational trends.

Because daily operational data often contains short-term fluctuations, the framework incorporates **rolling average analysis** using 7-day and 14-day windows to identify sustained pressure patterns. Capacity stress windows are defined as periods where net intake remains positive across multiple days while rolling system load exceeds the median load level. This approach allows early detection of prolonged operational strain that may otherwise remain hidden in daily reporting.

To support real-time monitoring and decision support, an **interactive dashboard** was developed using Streamlit and Plotly. The dashboard provides key performance indicators, rolling trend visualizations, net intake monitoring, and stress-window highlighting. These visual analytics tools help stakeholders quickly understand system behavior and detect emerging risks.

The analytical results demonstrate that sustained positive net intake leads to backlog accumulation within the care system. Rolling averages reveal pressure patterns that are not easily visible in raw daily data, while volatility spikes correspond to unstable operational periods. Time-series forecasting models, including ARIMA and Prophet, provide early warning signals for potential capacity strain and help anticipate future system demand.

Overall, this framework transforms operational datasets into a **scalable monitoring and forecasting system** that enhances situational awareness and supports proactive humanitarian response planning. Future improvements may include advanced seasonal forecasting models, automated model selection, integration of external event variables, and real-time alert systems.

Thank You.